

Restaurant Data Analysis and Machine Learning Project

Task 1: Restaurant Rating Prediction (Regression)

Objective

To predict the aggregate rating of restaurants based on features such as votes, price range, city, and service options.

Preprocessing

Handled missing values

Converted categorical variables using one-hot encoding

Converted binary features (Yes/No) into 1 and 0

Split dataset into training and testing sets (80:20)

Model Used

Decision Tree Regressor was used because it captures non-linear relationships and handles categorical data effectively.

Evaluation Metrics

MSE: 0.1146

RMSE: 0.3385

R² Score: 0.9496

Interpretation

The high R² value indicates that the model explains approximately 95% of the variance in restaurant ratings. Votes were found to be the most influential feature.

Task 2: Restaurant Recommendation System

Objective

To build a content-based recommendation system that suggests restaurants based on user preferences.

Features Used

Cuisine

City

Price range

Aggregate rating

Votes

Methodology

Applied one-hot encoding for categorical features

Used cosine similarity to compute similarity between restaurants

Developed a recommendation function that returns top 5 similar restaurants based on user input

Outcome

The system successfully recommends restaurants that match user preferences using similarity-based filtering.

Task 3: Cuisine Classification (Classification)

Objective

To classify restaurants into cuisine categories based on their attributes.

Preprocessing

Extracted first cuisine from multiple cuisine entries

Selected top 10 most frequent cuisines to reduce class imbalance

Converted categorical variables using encoding techniques

Model Used

Random Forest Classifier

Evaluation Metrics

Accuracy: 36.85%

Macro Average F1 Score: 0.20

Weighted Average F1 Score: 0.32

Interpretation

Cuisine classification is a multi-class problem with class imbalance. After reducing categories to the top 10 cuisines, model performance improved significantly compared to using all cuisine classes.

Conclusion

This project demonstrates the application of:

Regression for numerical prediction

Content-based filtering for recommendations

Classification for category prediction

Each task highlights a different machine learning technique applied to real-world restaurant data.